Report 1

Machine Learning Assignment

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Group 2 - Members

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Topic

Comparative Study of Machine Learning and Deep Learning for Hate Speech Classification

Introduction

The rapid expansion of social media platforms has transformed how people communicate and share information online. While these platforms enable open discussion and global connectivity, they have also become channels for the spread of harmful content such as hate speech and offensive language. Detecting such content manually is increasingly difficult due to the massive volume of user-generated text. In recent years, techniques from Machine Learning and Deep Learning have shown promising potential for automatically identifying toxic language in online communication.

This project aims to develop and evaluate models for detecting hate speech in social media comments. Specifically, we compare traditional machine learning approaches with deep learning-based models to analyze their effectiveness in text classification tasks.

Dataset

Dataset Description

We use the **Twitter Toxic Tweets** dataset from Kaggle for the hate speech / toxic language detection task. The dataset contains **31,962** real-world Twitter posts labeled for toxic and non-toxic content. It includes three main columns: **id** (unique identifier), **label** (binary target, where 0 indicates non-toxic and 1 indicates toxic), and **tweet** (raw tweet text). Therefore, the dataset is suitable for a supervised binary text classification problem.

Why This Dataset Is Suitable

This dataset is appropriate for our project because it focuses on short, real-world social media text, which is often noisy, informal, and highly variable in writing style. Compared with longer and more formal text sources, tweets usually contain abbreviations, slang, hashtags, mentions, misspellings, and expressive punctuation. These characteristics make toxic tweet detection more challenging, but also more realistic for practical machine learning applications in online content moderation. In addition, the dataset size is manageable for both traditional machine learning models and deep learning models, which supports the comparative nature of our project.

One notable characteristic of this dataset is that it is based on Twitter text, which differs from standard document datasets in several ways. Tweets are typically short and context-dependent, so important signals may be compressed into only a few words, hashtags, emojis, or offensive expressions. This creates challenges for feature extraction because the model must learn from sparse and noisy text. Moreover, because the labels are binary, the task is more focused than multi-class or multi-label toxicity detection, allowing the group to concentrate on building a clear comparison between baseline machine learning approaches and neural models.

Expected Challenges

From a machine learning perspective, we expect several challenges from this dataset. First, tweet text may require substantial preprocessing because social media language is often unstructured. Second, toxic meaning can depend on wording patterns, sarcasm, or identity-related terms, which may be difficult for simple bag-of-words features to capture. Third, model evaluation should go beyond raw accuracy, because in toxic content detection, misclassifying toxic tweets as non-toxic is an important practical error. Therefore, metrics such as precision, recall, and F1-score will be important in our later experiments.

Source

- Source: Kaggle

- Dataset: Twitter Toxic Tweets
- URL: `https://www.kaggle.com/datasets/umitka/twitter-toxic-tweets`

Planned Direction of the Group

In the next phase of the project, our group plans to first conduct a more complete exploratory data analysis (EDA) to better understand the structure and quality of the dataset. We will examine the number of samples, column types, missing values, duplicate tweets, class distribution, and the length distribution of tweets. We also plan to inspect representative examples of toxic and non-toxic tweets in order to understand typical language patterns and possible annotation difficulties.

After that, we will design a preprocessing workflow suitable for social media text. This may include lowercasing, removing URLs, mentions, and unnecessary symbols, while carefully deciding whether hashtags, emojis, and punctuation should be preserved because they may carry toxic intent. We also plan to experiment with tokenization and text normalization methods before moving to feature extraction and model training.

Our final objective is to build a comparative pipeline that includes both traditional machine learning and deep learning methods. This aligns with the project topic, which is a comparative study between the two modeling families for toxic content classification. The goal is not only to obtain a good classification result, but also to analyze the strengths and limitations of each approach in terms of performance, interpretability, computational cost, and suitability for short noisy text.

Proposed Pipeline

The proposed pipeline of our project consists of the following stages:

- **Stage 1: Data understanding and inspection.** We will load the dataset, verify the columns and labels, inspect basic statistics, and identify possible issues such as duplicates, empty tweets, or text anomalies.
- **Stage 2: Data preprocessing.** We will clean the tweet text by handling lowercase conversion, links, usernames, special characters, and token normalization. At this stage, we will also compare light preprocessing and stronger preprocessing to examine whether aggressive cleaning removes useful toxic cues.
- **Stage 3: Exploratory data analysis.** We will analyze class balance, tweet length distribution, frequent words, and representative toxic expressions. Word clouds or top-frequency token comparisons between classes may be included to support interpretation.

- **Stage 4: Feature extraction.** For traditional machine learning models, we plan to use text vectorization methods such as Bag-of-Words and TF-IDF. For deep learning models, the input may be transformed into sequences of token indices and then mapped into embedding vectors.
- **Stage 5: Model development.** For machine learning baselines, we plan to experiment with models such as Naive Bayes, Logistic Regression, and Support Vector Machine. For deep learning, we may test models such as LSTM or another neural text classifier suitable for short sequences.
- **Stage 6: Evaluation and comparison.** We will divide the data into training and testing sets and compare models using accuracy, precision, recall, F1-score, and confusion matrix. The comparison will focus not only on which model performs best, but also on why certain models are more suitable for this type of text.
- **Stage 7: Discussion and conclusion.** Finally, we will summarize the main findings, discuss model limitations, and suggest possible improvements such as better preprocessing, pretrained embeddings, or transformer-based methods for future work.

Current Progress and Next Steps

At the current stage, our group has finalized the project topic and selected the Twitter Toxic Tweets dataset from Kaggle as the main data source. We have identified the task as a binary toxic tweet classification problem and started reviewing possible approaches from both machine learning and deep learning perspectives. The next step is to perform a more detailed exploratory data analysis and establish a preprocessing strategy tailored to noisy social media text. After that, the group will implement baseline traditional machine learning models, followed by neural models, in order to conduct a systematic comparison in the later stage of the project.